Elkarmi Amine



Final-Year Computer Engineering Student
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Profile

Final-year Computer Engineering student focused on AI, Data Science and Software Engineering, seeking a final-year internship (PFE). Interested in AI research, with the intention of pursuing a PhD in Artificial Intelligence after the engineering degree.

Education

Engineering Degree in Computer Engineering — 5th Year, EMSI 2026–2027
Integrated Preparatory Cycle — EMSI | **Baccalaureate in Physical Sciences** — Lycée Al Manal

Technical Skills — Project Exposure

AI & Data: PyTorch, scikit-learn, pandas, NumPy, Transformers, XGBoost, LightGBM, CatBoost, SHAP, Optuna.
Programming: Python, Java, C#, C++, JavaScript, TypeScript, PHP, Dart.
Backend: FastAPI, Flask, Django REST Framework, Spring Boot, ASP.NET Core, Node.js/Express.
Web & Mobile: React, Flutter; **Data Stores:** MongoDB, MySQL, PostgreSQL, OpenSearch.
Tools & Systems: Docker Compose, Git, Maven, gRPC, Streamlit, Ollama, ROS2, PX4, Gazebo.

Experience

PFA Internship — CLV TECH 1 July–31 August 2026
Hospital ERP: Request Management and Conversational AI

- Developed internal and patient request workflows with Django and PostgreSQL, including role-based access, audit trails, notifications and ReportLab PDF reports.
- Integrated an assistant using Qwen2.5:7B via Ollama with a Groq fallback, extracting structured intents and entities with user confirmation before database changes.

Third-Year Internship — Commercial Management Applications GitHub
Python, Flask, React, TypeScript, MongoDB, ttkbootstrap

- Web and desktop applications for products, orders and installation tracking, with an LLM assistant using catalog data as context through OpenRouter/DeepSeek.

Selected Projects

Federated Learning System GitHub
Python, PyTorch, FastAPI, Docker Compose, Streamlit

- Client–server system for distributed image classification: PyTorch clients train locally and send model updates for sample-weighted FedAvg aggregation.
- FastAPI coordinates rounds; Docker Compose runs the services and Streamlit monitors loss/accuracy. Includes ResNet, VGG and EfficientNet model support.

IT Project Cost and Duration Estimation GitHub
Python, scikit-learn, XGBoost, LightGBM, CatBoost, Optuna, SHAP, Streamlit

- Cost and duration prediction from project specifications, with feature engineering, cross-validation, Optuna tuning and stacked boosting regressors.
- Streamlit interface for document input and feature review; SHAP explanations and Plotly charts make prediction drivers visible.

Fake News Detection Platform — PFA25 GitHub
Python, FastAPI, Flask, Hugging Face Transformers, BERT, BART, OpenSearch

- Separate FastAPI service exposes pretrained BERT fake-news classification, BART article summarization and keyword extraction.
- Flask application backend integrates article management and OpenSearch retrieval; React provides the web interface.

HrayfiConnect — Artisan Services Platform GitHub
Flutter/Dart, Python, FastAPI, MongoDB/Motor, WebSockets, JWT, Cloudinary

- Flutter client connects to a FastAPI backend for artisan discovery, service bookings, user profiles, reviews and support tickets.
- MongoDB/Motor stores application data; WebSockets enable chat, JWT supports authentication and Cloudinary handles media uploads.

Autonomous Drone Inspection — Prototype GitHub
Python, ROS2, PX4, Gazebo, Docker, PyTorch, YOLO interfaces

- Simulated inspection of earthen structures with PX4/Gazebo flight, ROS2 waypoint image capture and A* navigation tools.
- Detection interfaces support YOLO/VLM backends and confidence-triggered revisits; real inference and evaluation require model/data assets.

Coursera Certifications

Introduction to Machine Learning — Duke University
Python for Data Science, AI & Development — IBM
Introduction to Big Data — UC San Diego
Introduction to NoSQL Databases — IBM
Introduction to Cloud Computing — IBM
Agile Project Management — Google
Software Engineering & Project Management — HKUST
OOP in C++ — EPFL